

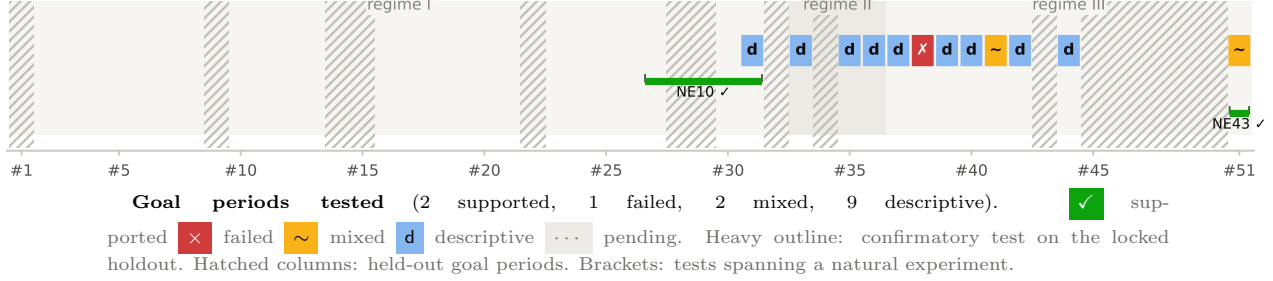
H35: The nudge is a measurably inefficient Maxwell demon

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 55% Faithfulness 2/5 Usefulness 2/5

Bottom line. The nudge is an inefficient Maxwell demon for activity (once-early nudging buys 2–3× more active minutes) and a worthless one for work (+0.22 commits/h, n.s.).

level: hypothesis · pipeline checks 86% · updated 2026-10-04



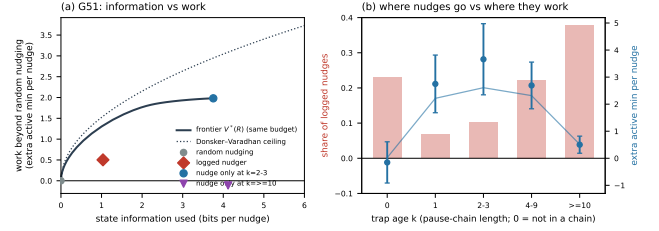
WHAT HAPPENED

H35 treats the auto-nudge as a feedback controller, i.e. a Maxwell demon. It measures the agent-state information in nudges, the activity they buy, and their distance from the best use of those bits. The test covers 13 non-holdout periods and two natural experiments (dated predictions; synthetic recovery checked first). G51 is the only well-powered period. There, a first nudge buys 1.45 [0.72, 2.20] extra active minutes and uses 1.4 bits of state. That is 38% of the work those bits could buy ($\eta_{SU} = 0.38 [-0.08, 0.68]$); only 15% of the bits are needed. Among pausing agents, the nudge does worse than random. It sends 38% of nudges to traps ≥ 10 pauses deep, which return 0.5 min each (2.2–2.6 min at $k = 1$ –9). Nudges add 0.5% of active minutes. Switch-on (NE10) and a silent stop on 08-20 do not show them. Under the old 12-h pause default, nudges wake agents at any trap age. **Round 1b** (work ledger, ledger calls): the nudge buys glances, not committed work or sustained runs. The holdout is not run.

PHYSICAL MODEL

Models 04 (Kolchinsky–Wolpert) and 02. The system X is an agent’s state: idle time, pause timing and trap age k (pause-chain length). The demon is the nudge $M \in \{0, 1\}$ at rate r . The work $g(x)$ is the extra active minutes in 30 min that a nudge causes in state x . The value of information ΔV is the work beyond random nudging. The frontier $V^*(R)$ is the most work that any policy with R bits can buy. s is the half-range of g and plays the role of $k_B T$. p_k is the probability of acting at the pause expiry.

$$\begin{aligned}\Delta V(\pi) &= \mathbb{E}[\pi g] - r \mathbb{E}[g], & I &= I(X; M) \\ V^*(R) &= \max_{I \leq R} \Delta V \Rightarrow \pi_\beta = \sigma(\logit r + \beta(g - \lambda)) \\ \eta_{SU} &= \Delta V_{\log} / V^*(I_{\log}), & \eta_{KW} &= R^*(\Delta V_{\log}) / I_{\log} \\ \Delta V &\leq s \sqrt{2rI} \quad (\text{Donsker–Varadhan}) \\ \logit p_k &= \alpha_i + \theta \ln k + (\gamma + \lambda \ln k) \mathbb{I}[\text{nudge}]\end{aligned}$$



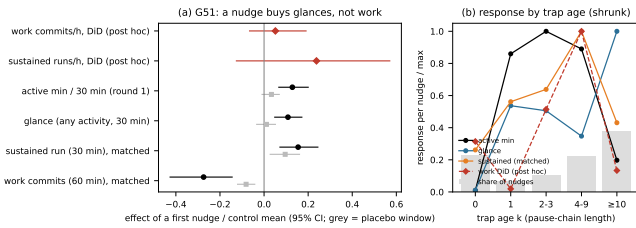
Observables. (a) G51 in trap-age space: work beyond random nudging against the state information used. The curve is the best that any policy can do with that many bits (dotted: analytic ceiling). The logged nudge sits far below it. Nudging only at $k = 2$ –3 reaches the end of the curve; nudging only deep traps is worse than random. (b) The nudge’s mass (bars) sits where the response per nudge (points) is lowest.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An operator who runs an idle-nudge can act on this. Nudge each pause chain once, at the first or second re-pause, and only while the agent is paused. Stop re-nudging agents stuck in long chains: they barely respond and need a reset or a new task. In G51 this policy gives $\times 2.3$ active minutes per nudge at the same budget. It does not buy committed work. The nudge buys only about 0.5% of agent activity, so even a perfect policy moves the swarm little. The thermodynamic frame works as accounting (frontier, ceiling, efficiency), not as physics. The policy result rests on one short-pause period.

RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b (G51)	Verdict
P1 3–8 bits	1.42 bits/nudge (unchanged)	partly
P2 work 0.5–3 min	1.45 active min → glance +0.08; commits +0.22/h [−0.29, 0.82] (DiD)	min yes; work no
P3 falls with trap age	minutes yes; sustained flat	minutes only
P4 $\eta_{SU} \leq 0.5$	0.38 (min) → glance 0.88; work undefined	activity only
P5 once early $\geq 1.25\times$	$\times 2.3$ min; work $\times 1.3$ (CI spans 0)	min only
NE43 (native)	work $-0.03 / -0.49$ day-SD at the two steps	yes
NE10 (native)	work $z = +0.92$	yes
G38 vs G51 (native)	sustained 0.058 vs 0.094	no



Second figure. Round 1b, G51. (a) Effect of a first nudge relative to the control mean (grey: placebo window before the nudge). Glances and active minutes rise, with clean placebos. Sustained runs and work commits fail the placebo, because nudged agents had just finished a burst. After differencing (red), both are zero. (b) Response by trap age. Active minutes peak at $k = 2-9$, glances peak in deep traps, and sustained work is flat.

GOAL PERIODS IN DETAIL

G31 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G33 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G35 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G36 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G37 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G38 $\times$ 1b: failed (native N1 as worded: sustained ATT 0.058 ...
G39 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G40 d 1b: descriptive (i 6 strictly isolated first nudges; ...
G41 $\sim$ 1b: mixed (round-1 statistics unchanged; no strictly ...
G42 d 1b: descriptive (i 6 strictly isolated first nudges; ...
+4 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Only G51 has power. The past-only design fails its placebo for sustained runs and work commits, so those numbers rest on a post hoc difference-in-differences. The once-early policy gain is in active minutes; for commits it is not identified. The identification assumes that the nudger acts on what the logs show; agent identity carries extra information. The 08-20 stop is undocumented. The card discloses three round-1 amendments and one round-1b post hoc step.

ROUND 2

Re-freeze `confirm.py` with glance and work outcomes. Then run it on #47 and #50 (C2–C4) and on #45 (wake-up). Test the once-early rule as a real intervention if operators allow it. Add an LLM-state feature (Jev behavior labels) to measure what the nudger reads beyond trap age.

No round-2 redirects yet.

H35-nudger-maxwell-demon · generated 2026-10-07 17:50 by infra/summaries/build_summaries.py · card hypotheses/H35-nudger-maxwell-demon/README.md · ratings: H35 round-1b agent (suggested) (page 2)